

Vector Integration: Green's Theorem in the Plane

1. Green's Theorem

Name of Concept

Green's Theorem in the Plane (Converting a Line Integral to a Double Integral).

Core Mathematical Formula

$$\oint_C (P dx + Q dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

Beginner-Friendly Explanation of Operations

- Why Use Green's Theorem?** Evaluating a line integral along a closed path with multiple straight segments (like a triangle or rectangle) usually forces you to calculate separate line integrals for each individual side. Green's Theorem is a major shortcut: it translates a difficult multi-stage loop integral along boundary C into a single double integral over the flat 2D region R trapped inside that loop.
 - Identifying P and Q :** Look closely at your line integral. The function multiplied by dx is always labeled P , and the function multiplied by dy is always labeled Q .
 - The Cross-Derivative Trick:** Inside the double integral, the order of differentiation is swapped:
 - You differentiate Q (the dy term) with respect to x .
 - You differentiate P (the dx term) with respect to y .
 - You subtract the second result from the first: $\left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right)$.
 - Setting Region Limits:** Once the components are subtracted, you integrate the remaining scalar function over the coordinate boundaries (x limits and y limits) that trace out the geometry of the trapped region R .
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2. Step-by-Step Problem Solution

Question

Evaluate by Green's Theorem $\oint_C (e^{-x} \sin y dx + e^{-x} \cos y dy)$ where C is the rectangle whose vertices are located at $(0, 0)$, $(\pi, 0)$, $(\pi, \frac{\pi}{2})$, and $(0, \frac{\pi}{2})$.

Step 1: Extract the components P and Q

Match the problem equation to the standard template form $(P dx + Q dy)$:

$$P = e^{-x} \sin y$$

$$Q = e^{-x} \cos y$$

Step 2: Compute the required partial derivatives

Differentiate P with respect to y , treating x as a constant:

$$\frac{\partial P}{\partial y} = e^{-x} \cos y$$

Differentiate Q with respect to x , treating y as a constant:

$$\frac{\partial Q}{\partial x} = -e^{-x} \cos y$$

Step 3: Setup the subtraction term for the double integrand

Substitute the computed derivative terms into the Green's Theorem component template:

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = (-e^{-x} \cos y) - (e^{-x} \cos y)$$

Combine the matching algebraic items:

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = -2e^{-x} \cos y$$

Step 4: Determine the integration limits from the rectangular vertices

Analyze the four corners of the rectangular boundary region R :

- The horizontal span runs along the x-axis from $x = 0$ to $x = \pi$.
- The vertical span runs along the y-axis from $y = 0$ to $y = \frac{\pi}{2}$.

Step 5: Assemble the complete double integral expression

Substitute the limits and the simplified internal function into the theorem layout:

$$\oint_C (P dx + Q dy) = \int_0^\pi \int_0^{\frac{\pi}{2}} (-2e^{-x} \cos y) dy dx$$

Step 6: Perform the inner integration with respect to y

Pull the terms depending purely on x outside the inner integration loop. Integrate $\cos y$ to get $\sin y$:

$$\int_0^{\frac{\pi}{2}} -2e^{-x} \cos y dy = -2e^{-x} [\sin y]_0^{\frac{\pi}{2}}$$

Evaluate the boundaries at $\frac{\pi}{2}$ and 0:

$$-2e^{-x} \left(\sin\left(\frac{\pi}{2}\right) - \sin(0) \right) = -2e^{-x}(1 - 0) = -2e^{-x}$$

Step 7: Perform the outer integration with respect to x

Place the result of the inner calculation into the outer integral loop:

$$\oint_C (P dx + Q dy) = \int_0^\pi -2e^{-x} dx$$

Apply standard exponential integration rules ($\int e^{-x} dx = -e^{-x}$):

$$\oint_C (P dx + Q dy) = [2e^{-x}]_0^\pi$$

Step 8: Evaluate the final numeric boundaries

Substitute the upper limit (π) and subtract the lower limit (0):

$$\oint_C (P dx + Q dy) = 2e^{-\pi} - 2e^0$$

Since any base raised to the power of zero equals 1 ($e^0 = 1$):

$$\oint_C (P \, dx + Q \, dy) = 2e^{-\pi} - 2(1)$$

Factor out the common scalar multiplier 2:

$$\oint_C (P \, dx + Q \, dy) = 2(e^{-\pi} - 1)$$